Powers of 10

Questions

1. a) Draw counters to represent **476** on a place value chart.

b) Draw counters to represent **4,760** on a place value chart.

c) Draw counters to represent **47,600** on a place value chart.

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2. Complete the sentences:

a) There are _____ ones in **50.**

b) There are _____ tens in **500.**

c) There are _____ hundreds in **5,000.**

d) There are _____ thousands in **50,000.**

→ **What pattern do you see as the numbers increase?**

3. Solve:

a) How many hundreds are there in **3,000**?

b) How many hundreds are there in **6,000**?

c) How many hundreds are there in **6,500**?

d) How many hundreds are there in **7,200**?

4. A truck's weight is measured in kilograms. Convert each weight into grams. (Remember: 1 kilogram = 1,000 grams)

a) How many grams are there in **5** kilograms?

b) How many grams are there in **500** kilograms?

c) How many grams are there in **2,000** kilograms?

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5. Use the Gattegno chart below to help answer the questions, 64,000 is shown in the chart.

100,000	200,000	300,000	400,000	500,000	600,000	700,000	800,000	900,000
10,000	20,000	30,000	40,000	50,000	60,000	70,000	80,000	90,000
1,000	2,000	3,000	4,000	5,000	6,000	7,000	8,000	9,000
100	200	300	400	500	600	700	800	900
10	20	30	40	50	60	70	80	90
1	2	3	4	5	6	7	8	9

a) Show **6,400** on the Gattegno chart.

b) Complete the sentences:

- **64,000** is _____ times the size of **6,400**.
- **640** is _____ times the size of **64**.
- **6,400** is one-tenth the size of _____.
- **640** is one-tenth the size of _____.

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6. Use the Gattegno chart again, 35,000 is shown in the chart.

100,000	200,000	300,000	400,000	500,000	600,000	700,000	800,000	900,000
10,000	20,000	30,000	40,000	50,000	60,000	70,000	80,000	90,000
1,000	2,000	3,000	4,000	5,000	6,000	7,000	8,000	9,000
100	200	300	400	500	600	700	800	900
10	20	30	40	50	60	70	80	90
1	2	3	4	5	6	7	8	9

a) Show **3,500** and **350** on the chart.

b) Complete:

- **35,000** is _____ times the size of **3,500.**
- _____ is one-hundredth the size of **3,500.**

7. A long rope measures several different lengths. Convert each length into centimetres.

a) How many centimetres are there in **6 metres**?

b) How many centimetres are there in **600 metres**?

c) How many centimetres are there in **6,000 metres**?

d) How many centimetres are there in **6,600 metres**?

(Remember: 1 metre = 100 centimetres)

8. The children pick number cards:

36,000	360	3,600	360,000

Tommy: My number is ten times bigger than Annie's.

Annie: My number has three hundreds.

Whitney: Dexter's number is one-tenth the size of my number.

Dexter: My number is one hundred times bigger than Annie's.

→ **Which number does each child have?**

9. Complete the pattern:

7,000 → **700** → **70** → ___ → ___

→ What operation is happening each time?

10. Lia says,

"When I divide 6,000 by 10, the answer is 60."

→ Is Lia correct?

→ Explain your answer.

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